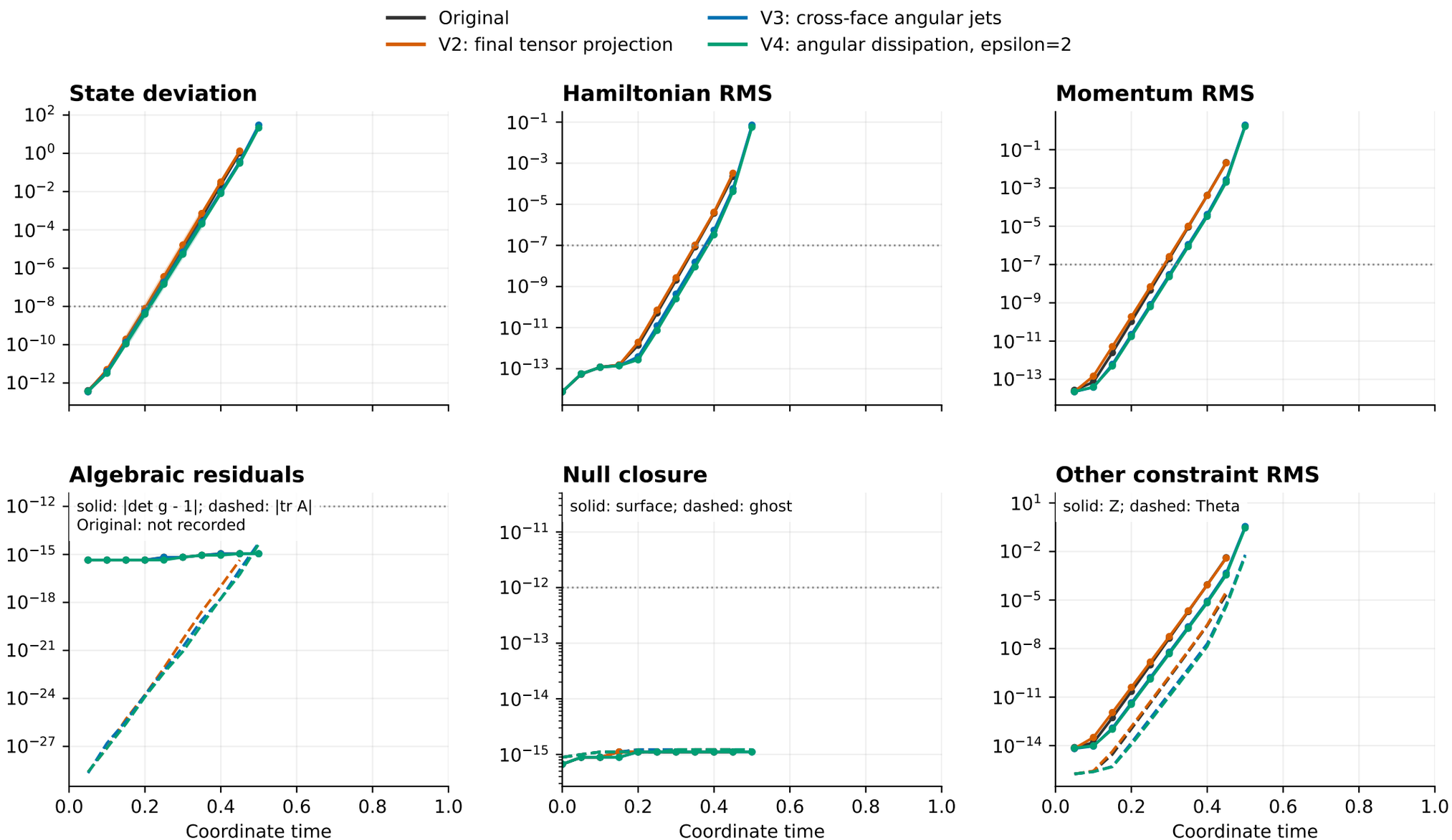

original: numerical failure; last diagnostic 0.45, accepted endpoint 0.4895
v2: numerical failure; last diagnostic 0.45, accepted endpoint 0.485
v3: numerical failure; last diagnostic 0.5, accepted endpoint 0.5205
v4: numerical failure; last diagnostic 0.5, accepted endpoint 0.5225

Dotted lines: declared limits. Curves stop at recorded diagnostics; no extension to later failure endpoints. Null precision alone is not stability.